DATA & TELECOMMUNICATION

Practical viva preparation | Semester 5

300 questions and answers

100 basic • 140 intermediate • 60 expert

10 practical topics across both uploaded lab notebooks

Prepared from your two DT practical files, six chapter-note PDFs and lecture slides, with worked examples and technical corrections. The focus is explaining, demonstrating and troubleshooting the experiments you may be assigned.

This is a broad preparation bank, not a promise of every possible question or a prediction of your examiner's choices. Practice values are labelled; unclear handwritten pin instructions are not treated as verified wiring.

A practical plan for tonight

Time	Focus
15 min	Read the scope and corrections; identify the software and demonstrated circuits.
55 min	ASK, FSK, BPSK, MUX and DEMUX: explain input, process and output.
45 min	Hamming: encode 1011, inject an error, calculate syndrome and correct it.
35 min	RS-232 and null modem: trace signal direction and diagnose failed transfer.
25 min	Topology and addressing: configure one LAN, ping it, explain switch loops.
45 min	Revise the question bank by weak topics, then rehearse the oral prompts.

If you only have one hour: read the correction page, the ten practical cards and the last-minute tables. Rehearse the aim, components, key rule, expected output and one fault for every practical.

Use the PDF bookmarks to jump between topics. The accompanying HTML contains the same 300 answers with search and level filters.

Scope: two notebooks, ten distinct topics

The experiment numbers differ between the two uploaded practical files. Use titles when matching your assigned experiment. Page references below are PDF page positions, not handwritten page numbers.

Topic	2-1. DT.pdf	2-2. DT.pdf
ASK generation	Exp. 1, pp.1-3	Exp. 1, pp.2-6
FSK generation	Exp. 2, pp.4-6	Exp. 2, pp.7-11
BPSK generation	Exp. 3, pp.7-10	Exp. 3, pp.12-16
4-to-1 MUX	Exp. 4, pp.11-12	Exp. 4, pp.17-21
1-to-4 DEMUX	Exp. 5, pp.13-14	Exp. 5, pp.22-26
Hamming generation	Exp. 6, pp.15-16	Exp. 6, pp.27-30
Hamming + error detection	Exp. 7, pp.17-19	Exp. 7, pp.31-35
Network topologies	Exp. 8, pp.20-25	Not in the index
RS-232 communication	Exp. 9, pp.26-28	Exp. 8, pp.36-40
Null-modem communication	Not included	Exp. 9, pp.41-44

How the levels work

Basic: definitions, component roles and direct observations. Intermediate: equations, worked examples and practical tests. Expert: assumptions, failure cases and design tradeoffs. Each topic has five basic, seven intermediate and three expert questions.

The uploaded experiments are primarily circuit/simulator exercises, with network configuration for the topology practical. There is no programming language requirement established by these notes; learn the circuit and its verification before trying to memorize code.

Corrections to read before memorizing

FSK needs two distinct frequencies

One handwritten FSK procedure repeats 1000 for both sine sources. A working BFSK demonstration must use f0 != f1. Check the actual demonstrated generator settings; the 1 kHz/2 kHz example here is illustrative.

Component names matter

74LS04 is a hex inverter, CD4016 is a quad bilateral switch, and 74LS138 is a 3-to-8 decoder with active-low outputs. The notes sometimes call these a transistor, gate or encoder. Name the actual function.

BPSK circuit instructions are ambiguous

The notes identify a 741 and LF398-family device. LF398 is a sample-and-hold IC. The scans disagree on the grounded 741 input; a conventional inverting stage grounds the non-inverting input, pin 3, and applies input/feedback at pin 2. Verify the complete demonstrated schematic and device package before wiring.

A single-error decoder can miscorrect two errors

Ordinary Hamming(7,4) has distance 3. It can detect any two-bit error when used only for detection, but cannot distinguish that case from a correctable single error using the three-bit syndrome alone. Extended Hamming adds overall parity for SECDED.

RS-232 is not TTL and MAX232 is not a UART

A real RS-232 connection uses bipolar line levels with inverted data polarity relative to typical UART logic. MAX232 performs level conversion; it does not serialize characters or set the baud rate. The notes' blanket statement that serial is always slower than parallel is incorrect.

The topology drawing is not the access protocol

A chain of switches is not a shared-coax bus, and a ring of Ethernet switches is not token ring. Redundant Ethernet paths need loop prevention such as Spanning Tree.

Simulation screenshots do not certify physical hardware

Power rails, signal range, logic-family thresholds, capacitors, LED resistors and connector orientation still matter. A simulator may hide supply pins or tolerate a schematic that would fail on a breadboard.

More theory and numerical corrections

SNR has no watt unit

10 mW / 1 microW = 10,000 as a dimensionless ratio, or 40 dB. D_T 04 appends W to the ratio; omit that unit.

Shannon output units

For B=3000 Hz and linear SNR=3162, C is approximately 34,881 bit/s = 34.9 kbit/s. The numerical slide's intermediate kbps label is wrong.

Quantization is amplitude mapping

A quantizer maps a sample to a finite amplitude level. Encoding assigns bits to that level. Neither operation is inherently encryption or guaranteed bandwidth compression.

Delta modulation needs a rate comparison

One bit per sample does not automatically mean lower total bit rate than PCM, because delta modulation generally oversamples. Compare required bit/s at the desired fidelity.

Even minimum distance can be useful

A distance-4 code corrects one error, or detects up to three errors if used only for detection. Extended Hamming uses distance 4 for single correction plus double detection; its extra distance is not simply wasted.

TDM slide input-rate mismatch

D_T 13 describes 3 kbit/s inputs but its solution uses 1 kbit/s. For three bit-interleaved 3 kbit/s sources without overhead: output 9 kbit/s, frame time 1/3000 s, output slot time 1/9000 s.

SONET is standardized

The lecture list saying SONET has no interoperable standard is misleading. SONET/SDH are standardized transport hierarchies. Vendor features and implementations can still affect interoperability.

State assumptions in every numerical answer

Use linear SNR in Shannon; give a bit order for Hamming; state active-high/low for logic outputs; distinguish waveform amplitude, peak-to-peak value and DC offset; include framing overhead when calculating serial character rate.

Question-bank map

Section	Questions	Topic
01	1-15	Communication fundamentals
02	16-30	Signals, noise and channel capacity
03	31-45	Practical: ASK generation
04	46-60	Practical: FSK generation
05	61-75	Practical: BPSK generation
06	76-90	Practical: 4-to-1 multiplexer
07	91-105	Practical: 1-to-4 demultiplexer
08	106-120	Practical: Hamming-code generation
09	121-135	Practical: Hamming error detection and correction
10	136-150	Practical: RS-232 serial communication
11	151-165	Practical: null-modem communication
12	166-180	Practical: network topologies
13	181-195	IP addressing and subnetting
14	196-210	OSI, TCP/IP and network devices
15	211-225	Transmission media and impairments
16	226-240	Line coding and scrambling
17	241-255	Sampling, PCM and delta modulation
18	256-270	Multiplexing and transmission timing
19	271-285	Error control, flow control, ARQ and HDLC
20	286-300	Switching, legacy networks and expert synthesis

Practical cards come next, followed by the 300 questions, calculation tables and oral-rehearsal prompts. All examples use the conventions printed alongside them.

Practical card 01 | ASK generation

2-1 pp.1-3; 2-2 pp.2-6

Aim

Generate OOK/binary ASK and relate carrier amplitude to input bits.

Build / identify

Locate the sine carrier, binary source, switching stage, complement control if used, ground and output scope probe. The notebooks use CD4016 and 74LS04. Verify supply rails and legal signal range in the actual model.

Run in order

1. Show the carrier alone and check amplitude/frequency. 2. Hold data low and record the output. 3. Hold data high and record it. 4. Apply 1010, then 110010. 5. Display data and output together with triggering on the data.

Expected result

With 1=ON and 0=OFF, sine bursts appear in high-data intervals and disappear in low intervals. Carrier frequency within each burst is unchanged. Measure it from one full period.

Practice setup

For a clear ideal demonstration, use a 100 bit/s data pattern and 1 kHz carrier, giving ten cycles per bit. Set amplitudes and offsets to fit the actual switch model; these values are not claimed as the assigned lab settings.

Troubleshoot

Always ON: wrong control or wrong probe. Always OFF: missing carrier, invalid control or power. Clipped output: signal outside supply range. Unstable trace: unsuitable trigger/time base.

Say aloud

I use binary data to control a carrier path. The output amplitude follows the bit value, while carrier frequency stays fixed. I verify both constant states and a changing pattern against the input trace.

Practical card 02 | FSK generation

2-1 pp.4-6; 2-2 pp.7-11

Aim

Select one of two carrier frequencies according to binary data.

Build / identify

Identify f0 and f1 sine sources, two switch branches, complementary controls and the combined output. Record which data state selects each source. The notes use CD4016 switching and an inverter.

Run in order

1. Confirm f0 and f1 separately. 2. Set data=0 and measure output frequency. 3. Set data=1 and measure again. 4. Use a pattern containing repeated and alternating bits. 5. Verify equal nominal amplitude and different cycle spacing.

Practice setup

Example only: f0=1 kHz, f1=2 kHz, data rate=100 bit/s. Each bit lasts 10 ms, containing 10 or 20 cycles. The two sources must not both be set to the repeated 1000 value in the handwritten procedure.

Expected result

Dense cycles for the higher-frequency state and wider-spaced cycles for the lower state. A switched pair of independent oscillators may produce phase discontinuities; do not automatically call it continuous-phase FSK.

Troubleshoot

One tone only: source values equal or one branch never selected. Strange sum: both branches enabled. Unequal amplitudes: generator settings or path gain differ.

Say aloud

FSK changes frequency instead of amplitude. I hold the data at each state to measure both tones, then show that the binary sequence selects them correctly.

Practical card 03 | BPSK generation

2-1 pp.7-10; 2-2 pp.12-16

Aim

Generate carrier states separated by 180 degrees and select them with data.

Identify the demonstrated circuit

The notebook shows a 741 inverting stage, equal 10 kOhm input/feedback resistors, an LF398-family device, clock/data selection, sources and a scope. LF398 is a sample-and-hold device; verify the exact simulator model and its wiring rather than copying uncertain handwritten pin instructions.

Check the inverter first

For the conventional inverting stage, input reaches the inverting input through Rin and feedback comes from the output through Rf; the non-inverting input is referenced to ground. With equal resistors gain=-1. Check suitable dual supplies and the actual op-amp limits.

Run in order

1. Display carrier and inverted carrier. 2. Confirm equal magnitudes and opposite polarity. 3. Hold the binary control at each state and compare the final output with the reference. 4. Apply 10110. 5. Identify phase changes only where the data changes.

Expected result

Under 1=reference, 0=inverted mapping, one state is in phase and the other is 180 degrees out of phase. A continuous carrier trace can still be BPSK; amplitude gaps are not required.

Troubleshoot

Clipping: supply/headroom/gain. No phase change: fixed control or wrong path. Output holds a flat sample: inspect how the LF398 control and hold node are used in the supplied circuit. Recheck the demonstrated design before substituting devices.

Say aloud

BPSK changes the sign of the carrier. I compare the output with a phase reference, because looking only at amplitude cannot prove correct phase modulation.

Practical card 04 | 4-to-1 multiplexer

2-1 pp.11-12; 2-2 pp.17-21

Aim

Route one of D0-D3 to output Y using select bits S1,S0.

Logic construction

Make NOT(S1), NOT(S0). Form T0=D0 NOT(S1) NOT(S0), T1=D1 NOT(S1) S0, T2=D2 S1 NOT(S0), T3=D3 S1 S0. OR the four terms to obtain Y.

Components

Use four three-input AND functions, two inverters, one four-input OR function, input toggles and output probes. CD4073 has three AND gates per package, so four functions need a second package or equivalent logic.

Run in order

1. Label select significance clearly. 2. Select 00,01,10,11 in order. 3. At each selection toggle its matching data input. 4. Toggle unselected inputs while keeping the selected input fixed. 5. Record output after the gates settle.

Expected result

00 -> D0; 01 -> D1; 10 -> D2; 11 -> D3. Example: with data 1,0,1,0 and selection 10, output is 1.

Troubleshoot

01/10 reversed: select wires swapped. Multiple inputs affect output: wrong product terms. Nothing passes: missing supply/enable or final OR connection. Brief transition glitches do not necessarily mean the settled truth table is wrong.

Say aloud

The select bits enable exactly one product term, so only that data input reaches the OR output. I verify all four selection cases, not just one screenshot.

Practical card 05 | 1-to-4 demultiplexer

2-1 pp.13-14; 2-2 pp.22-26

Aim

Route data D to one of Y0-Y3 using S1,S0.

Logic construction

Connect D to one input of each three-input AND function. Combine the other inputs with the four select minterms: NOT(S1)NOT(S0), NOT(S1)S0, S1 NOT(S0), S1S0. Keep the four outputs separate.

Run in order

1. Set D=1. 2. Step selection through 00,01,10,11. 3. Confirm the selected output moves from Y0 to Y3. 4. Repeat with D=0 and confirm all outputs are zero. 5. Record the eight input cases.

Expected result

For active-high gates, unselected outputs stay zero. With D=1 and selection 10, only Y2 is one. With D=0 no output is one, even though a route remains selected.

Useful follow-up

Holding D=1 turns this into a 2-to-4 decoder. MUX and DEMUX can share a time-slotted data line only when their selection timing is coordinated.

Troubleshoot

Two outputs high: check select complements and output polarity. No response to D: D wire missing. A 74LS138-style decoder has active-low outputs and must not be interpreted as this active-high truth table.

Say aloud

A DEMUX is a data distributor. The selected output follows D; selection alone does not force it high. I test both data values at every select setting.

Practical card 06 | Hamming encoder

2-1 pp.15-16; 2-2 pp.27-30

Aim and convention

Encode four data bits using even-parity Hamming(7,4). Print positions 1-7 from left to right: P1 P2 D1 P4 D2 D3 D4.

Build

Use input toggles D1-D4. Compute P1=D1 XOR D2 XOR D4; P2=D1 XOR D3 XOR D4; P4=D2 XOR D3 XOR D4. Each three-input XOR can be built from two two-input XOR gates.

Observe

Route each data/parity bit to its numbered output probe. LEDs on physical hardware need suitable current limiting; simulator indicator symbols may model that differently.

Worked input

For D1D2D3D4=1011, parity bits P1,P2,P4 are 0,1,0. The transmitted codeword is 0110011.

Verify

Group 1,3,5,7: 0 XOR 1 XOR 0 XOR 1=0. Group 2,3,6,7: 1 XOR 1 XOR 1 XOR 1=0. Group 4,5,6,7: 0 XOR 0 XOR 1 XOR 1=0.

Run and troubleshoot

Test 0000,1111,1011,1001 and preferably all 16 inputs. If parity fails, inspect XOR wiring and bit order. Replacing XOR with OR breaks cases containing more than one high input.

Say aloud

Three parity bits protect overlapping groups selected by binary position numbers. Their receiver checks can identify one erroneous position among seven, while syndrome zero represents no detected error.

Practical card 07 | Hamming detector and corrector

2-1 pp.17-19; 2-2 pp.31-35

Aim

Generate checks from a received word, identify an injected single-bit error and recover the original data.

Build checks

S1=r1 XOR r3 XOR r5 XOR r7; S2=r2 XOR r3 XOR r6 XOR r7; S4=r4 XOR r5 XOR r6 XOR r7. Feed the binary syndrome S4S2S1 to error-location logic.

Decoder detail

For a 74LS138, enable the device correctly and account for active-low selected outputs. Invert a selected line if an active-high correction toggle is needed. Syndrome zero must not flip a valid codeword bit.

Demonstrate

Encode 1011 -> 0110011. Inject an error at position 5 using an XOR toggle: received 0110111. Checks produce S4S2S1=101, or decimal 5. Flip position 5; recover 0110011, then read data at positions 3,5,6,7 -> 1011.

Test thoroughly

Try no error, each data-bit error and each parity-bit error. Show the corrupted word, syndrome and corrected word separately so a bypassed error-injection wire cannot fool the demonstration.

Limit

Two errors can produce a misleading nonzero syndrome. For safe single-correction/double-detection behavior, add overall parity and use the extended-Hamming decision rule; ordinary seven-bit logic does not establish SECDED.

Say aloud

The syndrome locates the error only under the assumed error model. A nonzero syndrome is evidence of a failed check, not proof that exactly one bit is wrong.

Practical card 08 | RS-232 and MAX232

2-1 pp.26-28; 2-2 pp.36-40

Aim

Demonstrate serial transfer and level conversion between logic signaling and an RS-232 path.

Identify domains

Mark each wire as logic-level or RS-232-level. For MAX232 channel 1: T1IN(11) -> T1OUT(14) is transmit conversion; R1IN(13) -> R1OUT(12) is receive conversion. VCC is pin 16 and GND is pin 15 on the 16-pin package.

Power

Follow the exact manufacturer circuit for the charge-pump capacitors and supply bypass capacitor. The classic MAX232 uses a 5 V supply; capacitor choices differ across related devices. Do not confuse charge-pump nodes with external power inputs.

Reference loopback test

Logic terminal TX -> T1IN. Connect T1OUT to R1IN on the RS-232 side. R1OUT -> logic terminal RX. This is an independent functional test; the notebook also describes a logic-side 11-to-12 jumper, whose surrounding terminal directions must be traced separately.

Run

Set matching data rate, data bits, parity and stop bits; 9600 8N1 is an illustrative choice. Disable local echo, send a known string and verify it returns through the loop. Remove the loop to confirm the return stops.

Troubleshoot

Nothing: wrong direction, port, power or handshake. Gibberish: timing or polarity. Double text: local echo plus returned text. Never judge success from local echo alone.

Say aloud

The UART or terminal creates serial framing. MAX232 only changes electrical levels. I verify the transmit and receive paths and compare the returned message with the sent one.

Practical card 09 | Null-modem communication

2-2 pp.41-44; not in 2-1

Aim

Exchange serial messages between two DTE endpoints without intermediate modems.

Physical principle

For standard PC DE-9 DTE ports, connect 3 -> 2, 2 -> 3 and 5 -> 5. This is the simple three-wire arrangement; hardware handshaking requires an appropriate additional wiring plan.

Simulation setup

Identify the Proteus COMPIM model, virtual terminal, any MAX232 stage and the host serial application. If using virtual paired ports, create or select the installed pair and give one endpoint to each application. COM9 in the notebook is an example, not a universal port.

Configure

Use the demonstrated model settings, matching endpoint framing and appropriate physical/virtual COM parameters. For a simple three-wire test, use no flow control when permitted. Check that no other application owns the chosen port.

Run

Send A-to-B: DT TEST A. Confirm only the peer receives it. Send B-to-A: REPLY B. Confirm both directions and disable local echo if it could mask a fault.

Troubleshoot

Port not found: wrong port/driver. Port busy: another program owns it. Only one direction: one crossed path missing. No data with good wiring: handshake mismatch. Model labels: verify source/output and sink/input direction, not just TX/RX text.

Say aloud

A null modem crosses the necessary endpoint signals. Unlike a loopback, it tests communication between two different endpoints; bidirectional messages prove both paths.

Practical card 10 | Network topology and ping

2-1 pp.20-25; not in 2-2 index

Aim

Build and test the topology patterns in the notes, with correct IP addressing and an explanation of physical versus logical behavior.

Start with a star

Place a switch and four PCs in Packet Tracer. Connect each PC to a suitable switch port. Assign 192.168.10.1-.4 with mask 255.255.255.0. Use unique addresses and wait for links to become operational.

Verify

From PC1 test ping 192.168.10.2, then the other peers. Use the simple-PDU or simulation view to inspect ARP and ICMP. A default gateway is unnecessary for these same-subnet exchanges.

Extend

Construct the requested chain, ring, tree, mesh or hybrid using the lab device selection. For five nodes a complete undirected mesh has ten links. Label each physical link and explain which paths actually forward traffic.

Important distinction

A chain of switches is not a true shared-coax bus. A physical Ethernet ring is not token ring. In a redundant switched topology, STP can intentionally block some links to prevent loops.

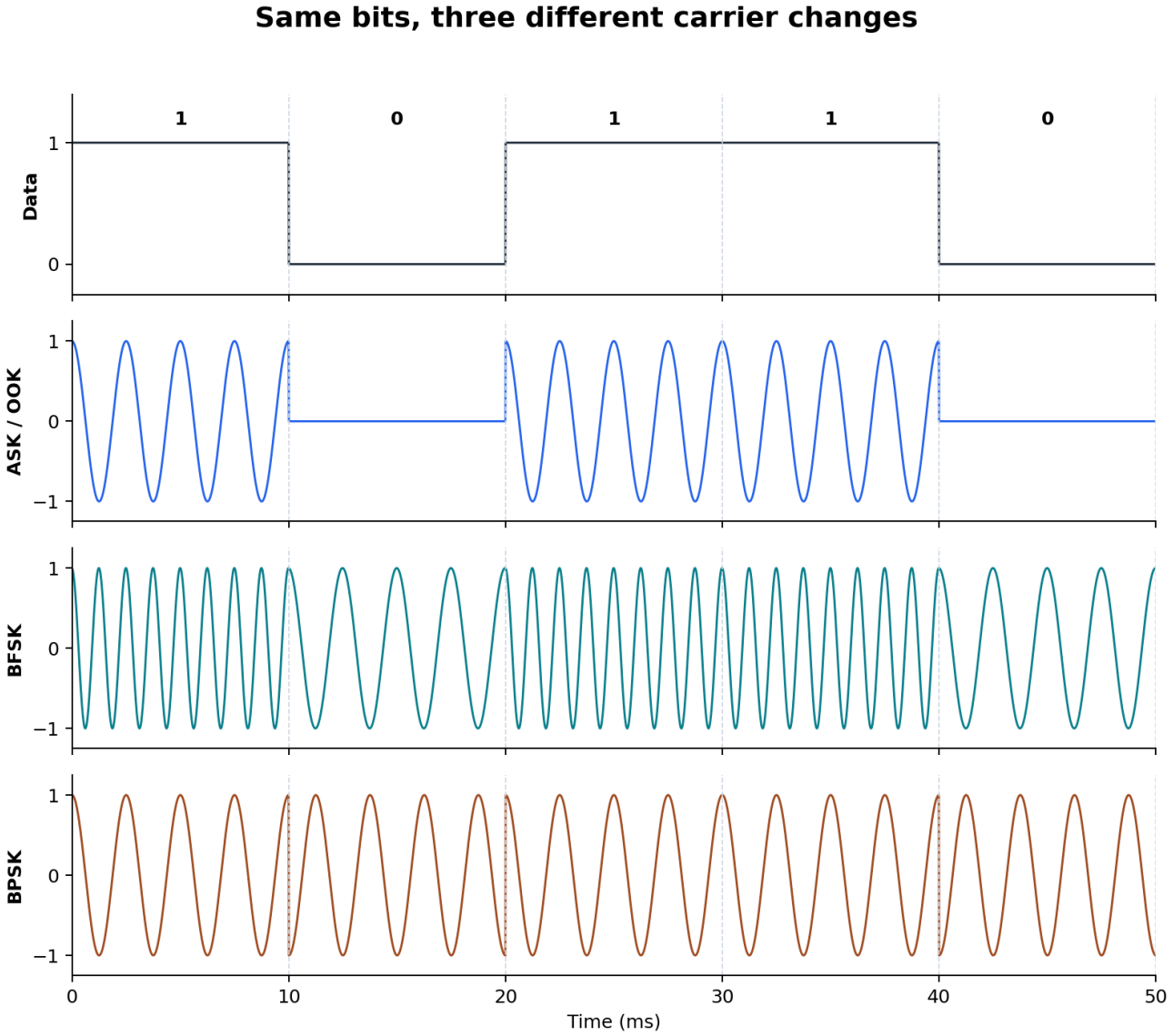
Troubleshoot

Check link state/cable, IP uniqueness, mask, VLAN membership and STP state. If the first ping fails, inspect address resolution and convergence, then retry. Crossing subnets additionally requires routing and correct gateways.

Say aloud

I verify the topology through connectivity tests, not only the diagram. The forwarding protocol and addressing determine communication; the physical shape alone does not.

Read the waveform, explain the difference



Original illustrative signals, not captured simulator results. Data rate=100 bit/s; ASK/BPSK carrier=400 Hz. BFSK uses 400 Hz for 0 and 800 Hz for 1, with continuous phase in this drawing. The lab's switched oscillators need not preserve phase.

Scheme	Changes with the data	Look for
ASK / OOK	Amplitude	Carrier present for 1, absent for 0
BFSK	Frequency	Different cycle density in each state
BPSK	Phase / sign	180-degree relation against a reference

BPSK mapping here: 1=reference, 0=negative reference. The two consecutive 1s keep the same phase state. A phase transition can occur smoothly at a zero crossing or show a jump at another carrier phase.

01 / Communication fundamentals

1. [Basic] What is data communication?

It is the exchange of data between devices through a transmission medium using agreed rules. A working system needs a sender, receiver, message, medium and protocol.

2. [Basic] What is telecommunication?

Communication over a distance using electrical, electromagnetic or optical signals. Data communication focuses on exchanging represented information; telecommunication also includes voice and video services.

3. [Basic] What makes communication effective?

Correct delivery, accuracy, timely arrival and acceptable jitter. A perfectly received voice packet can still be useless if it arrives too late for playback.

4. [Basic] What is a protocol?

A set of communication rules covering syntax, semantics and timing: how a message is formatted, what its fields mean, and when or how fast it is sent.

5. [Basic] What is the difference between data and a signal?

Data is the information; a signal is its physical representation. The same binary data can be represented by voltage pulses, optical pulses or a modulated radio carrier.

6. [Intermediate] Distinguish analog and digital signals.

An analog signal varies continuously over its allowed values; a digital representation uses discrete states. Real digital waveforms still have finite rise time, noise and analog electrical behavior.

7. [Intermediate] Explain simplex, half-duplex and full-duplex.

Simplex sends in one direction, half-duplex permits both directions at different times, and full-duplex permits simultaneous two-way communication. Broadcast reception, a walkie-talkie and a telephone conversation are examples.

8. [Intermediate] Are serial and simplex the same?

No. Serial means bits are sent sequentially on a channel. Separate transmit and receive paths can make a serial connection full-duplex.

9. [Intermediate] Distinguish bit rate and baud rate.

Bit rate counts bits per second; baud counts symbols per second. For fixed-length mapping with M distinct symbols, each symbol represents log2(M) bits, so gross bit rate equals symbol rate times log2(M).

10. [Intermediate] What are bandwidth, throughput and goodput?

Bandwidth may mean a frequency range in hertz or an informal link capacity in bit/s. Throughput is achieved transfer rate; goodput counts useful application data after overhead and retransmissions.

11. [Intermediate] What is jitter?

Variation in packet delay or signal timing. A playback buffer can absorb some delay variation, but increases latency and cannot recover data that misses its deadline.

12. [Intermediate] What does the phrase 'without coding' mean in these practicals?

The experiments are built using circuit components and simulator instruments, rather than a program. It does not mean that binary representation or error-control coding is absent: Hamming coding is explicitly one experiment.

13. [Expert] Can digital communication be completely noise-free?

No. Receivers make decisions about noisy waveforms. Regeneration and error control improve reliability, but sufficiently strong noise, timing errors or interference can still cause incorrect bits.

14. [Expert] Why does higher bit rate sometimes reduce reliability?

With unchanged power and channel conditions, shorter bit intervals may reduce energy per bit and increase sensitivity to bandwidth limits and timing errors. Modulation, coding and receiver design determine the actual tradeoff.

15. [Expert] How should you explain an unfamiliar assigned circuit?

Identify the input information, carrier or clock, processing blocks, output and measurement points. Then trace one simple input through it and state the observable test that would confirm its intended behavior.

02 / Signals, noise and channel capacity

16. [Basic] What are amplitude, frequency and phase?

Amplitude describes signal magnitude, frequency describes cycles per second, and phase describes position within a cycle relative to a reference. These are the carrier properties varied by ASK, FSK and PSK.

17. [Basic] What is the relation between frequency and period?

T = 1/f. A 1 kHz sinusoid has period 1 ms. Measure one complete cycle between equivalent points, such as two consecutive positive peaks.

18. [Basic] What is attenuation?

Loss of signal power during transmission. Attenuation is often expressed in decibels; amplifiers or repeaters may compensate for it, subject to noise and distortion limits.

19. [Basic] What is distortion?

A change in waveform shape, for example when frequency components experience unequal attenuation or delay. A signal may be distorted even when its average power remains adequate.

20. [Basic] What is noise?

Unwanted energy that interferes with the desired signal. Common examples are thermal noise, impulse noise, crosstalk and intermodulation noise.

21. [Intermediate] How do you calculate SNR?

SNR = signal power/noise power, a dimensionless ratio. SNR in dB = 10 log10(SNR). For 10 mW signal and 1 microW noise, the ratio is 10,000 and the result is 40 dB.

22. [Intermediate] When may 20 log10 be used?

For a voltage or current amplitude ratio when the impedances are equal, because power is proportional to amplitude squared. For a power ratio use 10 log10.

23. [Intermediate] State the Nyquist noiseless bit-rate relation.

For an ideal noiseless low-pass channel of bandwidth B and M signal levels, maximum bit rate is 2B log2(M). It assumes ideal pulse shaping; it is not a guarantee for a real noisy link.

24. [Intermediate] State Shannon capacity.

C = B log2(1 + SNR), with B in Hz and SNR as a linear power ratio. It gives a theoretical reliable-information-rate limit for the stated band-limited Gaussian-noise model.

25. [Intermediate] Find Nyquist rate for B = 3 kHz and four levels.

2 x 3000 x log2(4) = 12,000 bit/s. Four levels represent two bits per symbol, and the ideal maximum symbol rate is 6000 baud.

26. [Intermediate] Find Shannon capacity for B = 1 MHz and SNR = 63.

C = 10^6 x log2(64) = 6 Mbit/s. A practical design generally operates below this limit to achieve a useful error rate with realizable coding.

27. [Intermediate] Distinguish transmission and propagation delay.

Transmission delay is L/R for L bits at R bit/s. Propagation delay is distance/signal speed. Sending more bits increases transmission time, while a longer cable increases propagation time.

28. [Expert] Can increasing the number of signal levels beat Shannon capacity?

No. More levels increase the ideal bits per symbol, but make decisions more sensitive to noise. A reliable system must respect the channel's information capacity as well as implementation limits.

29. [Expert] What is wrong with a capacity result labelled 34,860 kbps for a 3 kHz channel with SNR 3162?

3000 x log2(3163) is about 34,881 bit/s, or 34.9 kbit/s. Multiplying hertz by a dimensionless logarithm yields bit/s, not kbit/s; the handwritten/slide unit is misleading.

30. [Expert] Why can a faster link still have high latency?

Increasing R reduces L/R but does not remove propagation, queueing or processing delay. A long-distance lightly loaded link can be propagation-limited even at a very high bit rate.

03 / Practical: ASK generation

31. [Basic] What is ASK?

Amplitude Shift Keying represents digital symbols with different carrier amplitudes. In binary ASK there are two amplitude states associated with the two binary values.

32. [Basic] What is OOK?

On-Off Keying is binary ASK with one amplitude equal to zero. With the mapping used here, 1 sends a carrier burst and 0 sends no carrier.

33. [Basic] What is the aim of the ASK experiment?

Generate an amplitude-keyed waveform from a sine-wave carrier and a binary control signal, then verify that the carrier appears or changes amplitude according to the data.

34. [Basic] Which instruments are used?

The notes show a sine generator, pulse/data generator, CD4016 switching circuit, inverter, resistors, reference ground and oscilloscope. Confirm the supplied simulator schematic and component power connections before running it.

35. [Basic] Which carrier property ideally stays unchanged in binary ASK?

Carrier frequency stays unchanged. The binary data changes amplitude; the phase reference is normally kept consistent across symbols when the source remains running.

36. [Intermediate] Write an OOK expression.

s(t) = A b(t) cos(2 pi fc t), where b(t) is 0 or 1. Thus s(t) is zero for a zero bit and a sinusoid of amplitude A for a one bit.

37. [Intermediate] What does the CD4016 do?

It provides four independently controlled bilateral analog switches. A control signal can connect or isolate a carrier path. It is not a four-input logic AND gate.

38. [Intermediate] What does the 74LS04 do?

It contains six digital inverters and can generate complementary control signals. It is not a single transistor and is not a linear sine-wave amplifier.

39. [Intermediate] How do you verify ASK on the oscilloscope?

Display the input data and output together. Trigger on a data edge and show several bit periods. With the declared mapping, high data should coincide with carrier bursts and low data with the off state.

40. [Intermediate] What test data is useful?

Begin with constant 0 and constant 1 to verify both states. Then use 1010 and a pattern with repeated bits, such as 110010, to check symbol boundaries and sustained states.

41. [Intermediate] Why should carrier frequency exceed data rate in a teaching display?

Several carrier cycles per bit make each burst easy to identify. This is a visibility choice for the demonstration, not a universal specification for every modulation system.

42. [Intermediate] Why might the output remain a sine wave continuously?

The switch may be permanently enabled, both controls may have the same state, the probe may be on the carrier instead of the output, or the binary generator may not be toggling.

43. [Expert] Why is ASK vulnerable to amplitude noise?

The receiver decides using amplitude. Noise, fading and gain variation can move the received amplitude across the decision threshold, especially when the two levels are poorly separated.

44. [Expert] Does an ideal OOK signal occupy just one frequency?

No. Keying changes the envelope and creates sidebands. Rectangular abrupt symbols have a broad spectrum; filtering or pulse shaping controls bandwidth and transition behavior.

45. [Expert] Why can a CD4016 simulation fail when connected to a bipolar sine source?

The analog voltage must stay within the switch's permitted supply range, and its control levels must be valid. A negative signal with a ground-referenced single supply may violate those limits; verify biasing, supplies and model compatibility.

04 / Practical: FSK generation

46. [Basic] What is FSK?

Frequency Shift Keying represents digital symbols by selecting different carrier frequencies. Binary FSK uses two tones, one for each bit value.

47. [Basic] What are mark and space frequencies?

They are the tones assigned to binary 1 and binary 0 respectively. State your chosen mapping; the higher frequency is not inherently required to represent 1.

48. [Basic] What is the aim of the FSK experiment?

Use binary data to select between two distinct sine-wave frequencies and observe the change in cycle spacing while keeping nominal amplitude similar.

49. [Basic] What components appear in the notes?

Two sine generators, a pulse generator, CD4016 analog switching, complementary control using an inverter, resistors and an oscilloscope.

50. [Basic] How does an FSK trace differ from ASK?

In FSK, cycle density changes between bits while nominal amplitude remains similar. In OOK, the carrier is present or absent while its frequency during bursts stays the same.

51. [Intermediate] Write the binary FSK signal rule.

For bit 0 transmit A cos(2 pi f0 t + phi0); for bit 1 transmit A cos(2 pi f1 t + phi1), where f0 differs from f1. Phase continuity depends on the implementation.

52. [Intermediate] How do two switches generate FSK?

One branch passes tone f0 and the other passes tone f1. Complementary data controls select one branch at a time so the output follows the intended tone.

53. [Intermediate] What happens if both generators have the same frequency?

Frequency no longer distinguishes the symbols. Any observed phase or amplitude change would be a switching artifact rather than the intended two-tone FSK behavior.

54. [Intermediate] Give a clear illustrative setup.

Use f0 = 1 kHz, f1 = 2 kHz, equal amplitudes and a data rate of 100 bit/s. Each 10 ms bit then contains 10 or 20 cycles. These are practice values, not prescribed lab settings.

55. [Intermediate] How do you measure the two output frequencies?

Hold the data at 0, measure output period T0, and calculate 1/T0. Repeat at 1 to obtain 1/T1. Then use alternating data to verify switching between those measured tones.

56. [Intermediate] Why are complementary controls needed?

They prevent both carrier branches from being selected together in the intended logical operation. Delay mismatch can still cause a brief overlap or gap in real switching hardware.

57. [Intermediate] How can BFSK be demodulated?

A receiver can compare energy in filters or correlators centered on f0 and f1. A suitable frequency discriminator is another approach. The experiment generates the signal; demodulation is a follow-up theory topic.

58. [Expert] Is FSK automatically continuous-phase?

No. Switching between independent oscillators can create phase jumps. Continuous-phase FSK uses a phase-continuous oscillator or an equivalent construction, not arbitrary independent-tone switching.

59. [Expert] What determines FSK bandwidth?

Tone separation, symbol rate, pulse shaping and the bandwidth measurement convention. There is no single exact bandwidth formula for every FSK implementation.

60. [Expert] Why does increasing tone separation have a tradeoff?

It may make the tones easier to distinguish with a given receiver, but usually consumes more spectrum. Required separation also depends on symbol duration and coherent or noncoherent detection.

05 / Practical: BPSK generation

61. [Basic] What is BPSK?

Binary Phase Shift Keying maps two bit values to carrier phases separated by 180 degrees. The nominal carrier amplitude and frequency are unchanged.

62. [Basic] What is another name for BPSK?

Two-phase PSK or phase-reversal keying. The two constellation points lie on opposite sides of the origin.

63. [Basic] How many bits does one BPSK symbol carry?

One bit, since log2(2) = 1. Without added coding overhead, bit rate and symbol rate are equal.

64. [Basic] What is the aim of the BPSK experiment?

Produce opposite-phase carrier states selected by binary data and verify the phase relation against a reference carrier.

65. [Basic] What is the role of a unity-gain inverting amplifier?

It produces -v(t). For a sinusoid, multiplication by -1 is equivalent to a 180-degree phase reversal, with unchanged amplitude magnitude.

66. [Intermediate] Write a BPSK equation with a declared bit mapping.

s(t) = A[2b(t)-1]cos(2 pi fc t). Here b=1 gives the reference carrier and b=0 gives its negative. Reversing the bit-to-phase mapping is also valid if both ends agree.

67. [Intermediate] What is the gain of an inverting op-amp with equal input and feedback resistors?

Gain is -Rf/Rin = -1. The notes show 10 kOhm resistors; the negative sign represents inversion, not a negative amplitude magnitude.

68. [Intermediate] How do you identify phase reversal on a scope?

Compare output with the reference carrier. In one data state their peaks align; in the other, a positive peak aligns with a negative peak. Display the data trace as well.

69. [Intermediate] Does every bit boundary cause a 180-degree change?

No. In ordinary BPSK, the selected phase changes when the bit value changes. Consecutive identical bits keep the same phase state.

70. [Intermediate] What is the LF398 named in the circuit notes?

A sample-and-hold IC. Treat the supplied Proteus circuit as a particular implementation to verify; do not describe LF398 as a generic BPSK chip or assume its pins match an ordinary op-amp.

71. [Intermediate] Why might BPSK appear like an uninterrupted sine wave?

The data may be constant, the time scale may be unsuitable, or a phase change may occur near a zero crossing. Compare against the carrier and data rather than looking for an amplitude gap.

72. [Intermediate] How is coherent BPSK detected?

Multiply or correlate the received signal with a phase-aligned carrier, integrate over the symbol interval, and decide from the sign. Timing and carrier synchronization are essential.

73. [Expert] What is the 180-degree carrier ambiguity?

A recovered carrier can have the opposite sign, reversing all decoded bits. A known preamble or differential encoding can resolve the ambiguity.

74. [Expert] What is the difference between BPSK and differential BPSK?

Ordinary BPSK encodes a bit in an absolute phase state relative to a reference. Differential encoding represents information using the change between successive phase states; the encoding and detection method must be specified.

75. [Expert] Why is 'phase changes but frequency stays constant' an idealized statement?

Within a symbol the sinusoid has the carrier frequency. Abrupt phase transitions create spectral components and transition behavior; they do not mean the complete modulated signal is a single-frequency sinusoid.

06 / Practical: 4-to-1 multiplexer

76. [Basic] What is a multiplexer?

A combinational circuit that selects one of several data inputs and forwards it to one output. It is often called a data selector.

77. [Basic] How many select lines does a 4-to-1 MUX need?

Two, because two binary select bits provide four combinations. In general, a 2^n-to-1 MUX needs n select bits.

78. [Basic] State the selection table.

With S1 as the most significant select bit: 00 selects D0, 01 selects D1, 10 selects D2, and 11 selects D3.

79. [Basic] Which basic gates implement this MUX?

Four three-input AND functions, two NOT functions and one four-input OR function. Equivalent implementations can use other gate arrangements.

80. [Basic] Is a MUX sequential or combinational?

Combinational. Its ideal output depends on present data and select inputs; it does not need a clock or store past input values.

81. [Intermediate] Write the Boolean output equation.

Y = D0 NOT(S1) NOT(S0) OR D1 NOT(S1) S0 OR D2 S1 NOT(S0) OR D3 S1 S0. Only the selected product term is enabled.

82. [Intermediate] What output results for D0,D1,D2,D3 = 1,0,1,0 and S1S0 = 10?

D2 is selected, so Y = 1. The unselected inputs do not affect the ideal settled output.

83. [Intermediate] Why are inverted select signals required?

Each input must be enabled by one unique select combination. Complemented signals produce the combinations containing a zero select bit.

84. [Intermediate] How do you test a MUX beyond one example?

For each of the four select settings, toggle the selected data input and verify the output follows it. Toggle unselected inputs and confirm they do not change the settled output.

85. [Intermediate] What mistake occurs when S0 and S1 are swapped?

The meanings of select codes 01 and 10 exchange. State which wire is the most significant bit and check the wiring against the equation.

86. [Intermediate] How can a 4-to-1 MUX implement XOR of two variables?

Use A,B as select bits and set D0=0, D1=1, D2=1, D3=0. The selected constants reproduce the XOR truth table.

87. [Intermediate] How can you build an 8-to-1 MUX using 4-to-1 blocks?

Use two first-stage 4-to-1 MUXes controlled by the lower two select bits, then a 2-to-1 stage controlled by the third bit. A third 4-to-1 device can be wired to act as that 2-to-1 stage.

88. [Expert] Why may a physical MUX briefly glitch when select bits change?

Different propagation delays can temporarily enable the wrong product term or none. Register the output or sample after settling when a glitch would matter.

89. [Expert] Is one CD4073 enough for four three-input AND functions?

No. A CD4073 contains three three-input AND gates, so four such functions require more than one package or a different implementation. A simulator can hide this package-level detail.

90. [Expert] Is the logic MUX experiment the same as FDM?

No. The circuit selects a data input at an instant. FDM simultaneously allocates separate frequency bands to channels. A periodically controlled MUX can be part of a time-division system, but the bare circuit does not implement FDM.

07 / Practical: 1-to-4 demultiplexer

91. [Basic] What is a demultiplexer?

A combinational circuit that routes one data input to one of several outputs, as selected by control bits. It is a data distributor.

92. [Basic] How many lines does a 1-to-4 DEMUX have?

One data input, two select inputs and four data outputs, excluding power and any separate enable input.

93. [Basic] What happens to unselected active-high outputs?

They remain logic 0. The selected output follows the data input; it is not necessarily 1.

94. [Basic] What happens when D = 0?

All four outputs are 0 in the active-high AND-gate implementation, regardless of the select bits.

95. [Basic] Which gates implement the lab DEMUX?

Four three-input AND functions and two inverters for the select signals. No final OR gate is needed because the four outputs remain separate.

96. [Intermediate] State the output equations.

Y0 = D NOT(S1) NOT(S0); Y1 = D NOT(S1) S0; Y2 = D S1 NOT(S0); Y3 = D S1 S0.

97. [Intermediate] What happens for D = 1 and S1S0 = 10?

Y2 = 1 and Y0,Y1,Y3 = 0. If D then becomes 0, all outputs become 0.

98. [Intermediate] How do you verify the whole circuit?

For each select combination test D=0 and D=1, giving eight cases. Record Y0 through Y3 and compare with the equations.

99. [Intermediate] How does a DEMUX differ from a decoder?

A decoder selects an output from a binary code. A DEMUX additionally routes a data value; a decoder with an appropriate enable can perform a DEMUX function.

100. [Intermediate] How can a 1-to-4 DEMUX behave as a 2-to-4 decoder?

Hold its data input at 1. Then each select combination activates one output, subject to the circuit's active-high or active-low convention.

101. [Intermediate] How can MUX and DEMUX transfer several channels over one data line?

The sender selects channels in sequence and the receiver routes successive time slots to corresponding outputs. They need synchronized selection and, usually, storage to retain each channel's value between slots.

102. [Intermediate] Why do two outputs sometimes appear active in simulation?

Check select complements, duplicate connections, whether outputs are active-low, and whether the display catches a switching transient. Hold the select bits steady before judging the truth table.

103. [Expert] Is a DEMUX simply a digital MUX wired backwards?

Not generally. Logic-gate implementations have directional signal flow. A bilateral analog switch network may support both directions within its ratings, but a gate-based MUX does not automatically become a DEMUX when reversed.

104. [Expert] What does an active-low decoder change in the interpretation?

The selected output is 0 and unselected outputs are 1. Inversion may be needed before a circuit expecting an active-high one-hot correction signal.

105. [Expert] Why can routing data without a clock still create timing problems?

Select signals and data have propagation delays and may change together. The receiver needs a valid-data interval or sampling rule; combinational routing alone does not provide reliable timing synchronization.

08 / Practical: Hamming-code generation

106. [Basic] What is Hamming coding?

A family of linear error-correcting block codes that add parity constraints to data. The lab's common Hamming(7,4) example uses four data bits and three parity bits.

107. [Basic] What does (7,4) mean?

Each codeword contains seven bits, four carrying data and three providing redundancy. Its code rate is 4/7.

108. [Basic] What is a parity bit?

A check bit chosen so a selected group has even or odd total parity. It adds a consistency condition; it is not itself a copy of one specific data bit.

109. [Basic] Where are parity bits placed in this guide?

At positions 1,2,4, which are powers of two. Data D1,D2,D3,D4 occupy positions 3,5,6,7. Positions are printed left to right in increasing order.

110. [Basic] Which gate computes even-parity check bits?

XOR. XOR of the selected data bits gives a parity bit that makes the XOR of the complete protected group equal to zero.

111. [Intermediate] How many parity bits are needed for m data bits?

Choose the smallest r satisfying 2^r >= m+r+1 for single-error location, including the no-error case. For m=4, r=3 because 8 >= 8.

112. [Intermediate] What are the even-parity equations?

P1 = D1 XOR D2 XOR D4; P2 = D1 XOR D3 XOR D4; P4 = D2 XOR D3 XOR D4. Codeword order is P1,P2,D1,P4,D2,D3,D4.

113. [Intermediate] Why does P1 cover positions 1,3,5,7?

Those position numbers have least significant binary bit 1. P2 covers 2,3,6,7 and P4 covers 4,5,6,7 using the other binary position bits.

114. [Intermediate] Encode data D1D2D3D4 = 1011 with even parity.

P1=1 XOR 0 XOR 1=0; P2=1 XOR 1 XOR 1=1; P4=0 XOR 1 XOR 1=0. In increasing position order the codeword is 0110011.

115. [Intermediate] Encode data 1001 using the same convention.

P1=0, P2=0 and P4=1, so the codeword is 0011001. Do not compare this printed string with another source until its bit order and parity convention are known.

116. [Intermediate] How is the encoder built using two-input XOR gates?

Each parity is the XOR of three data bits, made by cascading two XOR gates. Three independent parity networks use six XOR gates; route the four data bits directly to their codeword positions.

117. [Intermediate] How do you test the encoder?

Apply several data patterns, read all seven bits and independently XOR each parity group. For full coverage, test all 16 possible four-bit input words.

118. [Expert] Why must the bit-position convention be stated first?

Some diagrams number from the right, use D0-D3, or print parity bits separately. Equivalent encoders can produce different-looking strings; syndrome interpretation is only valid with a consistent mapping.

119. [Expert] Why are three checks sufficient for seven single-bit locations?

Three syndrome bits provide eight patterns: zero for no detected error and seven nonzero patterns for the seven positions. Each column of the parity-check matrix is a different nonzero three-bit position label.

120. [Expert] What is the cost of this error correction?

Redundancy reduces information rate: only four of every seven transmitted bits are data. It also adds encoder/decoder logic and may add latency, in exchange for correction of a single error per codeword.

09 / Practical: Hamming error detection and correction

121. [Basic] What is a syndrome?

The vector of parity-check results computed from the received word. For this guide write it as S4 S2 S1; interpret it as a binary position number under the single-error assumption.

122. [Basic] What does syndrome 000 mean?

All parity checks pass. It means no error is detected, not that arbitrary multiple-bit corruption is impossible.

123. [Basic] What does syndrome 101 mean?

Decimal position 5 is implicated if there is at most one error in the seven-bit codeword. Flip that bit to correct a single-bit error.

124. [Basic] What is Hamming distance?

The number of bit positions in which two equal-length words differ. For example, 1011 and 1001 differ in one position.

125. [Basic] What is the minimum distance of Hamming(7,4)?

Three. This supports correction of one arbitrary bit error, or detection of up to two errors when used only as an error detector.

126. [Intermediate] Write the syndrome equations.

S1 = r1 XOR r3 XOR r5 XOR r7; S2 = r2 XOR r3 XOR r6 XOR r7; S4 = r4 XOR r5 XOR r6 XOR r7. These equations use even parity and the stated numbering.

127. [Intermediate] Correct received word 0110111.

Starting from encoded 1011 = 0110011, position 5 has flipped. Checks give S1=1, S2=0, S4=1, so syndrome 101 indicates position 5. Flipping it restores 0110011 and data 1011.

128. [Intermediate] What happens if a parity bit itself is corrupted?

It is still a single-bit error and produces the corresponding position syndrome. For example, an error in position 4 gives syndrome 100; the correction flips that parity bit.

129. [Intermediate] How can a decoder and XOR gates correct the word?

Decode the nonzero syndrome into a one-hot error-location signal, then XOR that signal with the corresponding received bit. For syndrome zero no codeword bit should be flipped.

130. [Intermediate] What is the function of the 74LS138 in the notes?

It is a 3-to-8 decoder/demultiplexer, not an encoder. Its selected output is active-low, so correction logic must account for polarity and enable inputs.

131. [Intermediate] How should error injection be demonstrated?

Generate a valid word, XOR a chosen bit with an error toggle, and show the corrupted word, syndrome, corrected word and recovered data. Repeat for each position, not just one data bit.

132. [Intermediate] What extra bit produces extended Hamming SECDED?

An overall parity bit extends Hamming(7,4) to (8,4), with minimum distance four. It permits single-error correction and double-error detection under the assumed up-to-two-error model.

133. [Expert] Why can ordinary Hamming miscorrect two errors?

Two erroneous positions combine to produce a syndrome that also names another position. A decoder that assumes one error flips that third bit, potentially producing a different valid codeword.

134. [Expert] Give a two-error counterexample.

Starting from 0110011, flip positions 1 and 2 to get 1010011. Syndrome is 011, so a naive correction flips position 3 and produces 1000011, a wrong valid codeword. Detection and safe correction are different capabilities.

135. [Expert] State the extended-Hamming decision rule.

Let S be the three-bit syndrome and P the XOR of all eight received bits. With at most two errors: S=0,P=0 means no detected error; S!=0,P=1 means correct position S; S=0,P=1 means flip overall parity; S!=0,P=0 means detected double error, do not correct.

10 / Practical: RS-232 serial communication

136. [Basic] What is RS-232?

An electrical and interface standard for serial communication between equipment. It specifies signal behavior and interface conventions; it is not an application message format like a text-command language.

137. [Basic] What is a UART?

A Universal Asynchronous Receiver/Transmitter. It converts parallel bytes to serial frames and received frames back to bytes, handling start/stop timing and optional parity.

138. [Basic] What is the MAX232?

A line driver/receiver that translates between logic-level signals and RS-232 levels. It is not a UART and does not generate baud timing or interpret characters.

139. [Basic] What do TX and RX mean?

Transmit and receive, relative to the named device or endpoint. An output must ultimately drive the other endpoint's input; connector labels can be referenced to DTE conventions.

140. [Basic] What does 9600 8N1 mean?

A bit rate of 9600 bit/s, eight data bits, no parity and one stop bit. With one start bit there are ten transmitted bits per character.

141. [Intermediate] What are the data polarities on RS-232?

Mark, representing data 1 and idle, is negative; space, representing data 0, is positive. Receiver levels beyond -3 V and +3 V are valid respective states; the region between is undefined by the interface specification.

142. [Intermediate] Which parameters must agree at the two endpoints?

Data rate, data-bit count, parity, stop-bit configuration and flow-control expectations. Electrical compatibility and correct signal direction are also necessary.

143. [Intermediate] What is the maximum 8N1 character rate at 9600 bit/s?

9600/10 = 960 characters per second, assuming continuous transmission without extra gaps or flow-control pauses. Useful payload bit rate is 7680 bit/s.

144. [Intermediate] Explain the two MAX232 paths used in a loopback.

Logic input T1IN pin 11 drives RS-232 output T1OUT pin 14. RS-232 input R1IN pin 13 produces logic output R1OUT pin 12. Connecting 14 to 13 forms an external RS-232-side loopback.

145. [Intermediate] Why are capacitors used with MAX232?

They support its charge pump, which creates positive and negative line-driver supplies. Their values and connections must match the exact device datasheet; MAX232-family variants are not all identical.

146. [Intermediate] Why might random characters appear?

Check mismatched baud/framing, wrong polarity, invalid voltage levels, noise, missing reference ground or simulation timing. Use a repeated known character and scope the frame to narrow the fault.

147. [Intermediate] What is a framing error?

The receiver does not see the expected stop-bit condition at the expected time. Wrong bit timing, a break condition or corrupted signaling can cause it.

148. [Expert] Can a TTL UART connect directly to a real RS-232 port?

Not as a general electrical interface. Their voltage levels and polarity differ, and RS-232 voltages can damage logic inputs. Use an appropriate transceiver or the correct interface adapter.

149. [Expert] Does text visible with local echo prove transmission worked?

No. Local echo can display keystrokes before anything returns. Disable it or use a distinct remote reply to prove that the complete receive path works.

150. [Expert] Is serial always slower than parallel?

No. Serial links can run at high rates with fewer skew and crosstalk problems. Compare a specific serial standard and physical implementation rather than assuming the number of wires determines speed.

11 / Practical: null-modem communication

151. [Basic] What is a null-modem connection?

A direct connection between two DTE endpoints that crosses the necessary serial signals, allowing communication without modems between them.

152. [Basic] What are DTE and DCE?

Data Terminal Equipment is an endpoint such as a computer; Data Circuit-terminating Equipment provides the communication-side interface, traditionally a modem. Signal direction depends on which role a connector follows.

153. [Basic] What is the minimum three-wire DE-9 DTE-to-DTE null-modem wiring?

Cross pin 3 TX at one end to pin 2 RX at the other, cross the reverse direction, and connect signal ground pin 5 to pin 5. This assumes the standard PC-style DTE pin assignment.

154. [Basic] Why does a straight-through cable fail between two standard DTE ports?

It connects transmit to transmit and receive to receive. The null-modem crossover connects each transmitter to the opposite receiver.

155. [Basic] What is COMPIM used for in the lab?

It connects a Proteus simulation to a host computer serial-port interface. Its configured COM port must exist and be available to the simulator.

156. [Intermediate] Is COM9 a mandatory setting?

No. It is an example in the notes. Use the actual available port or one endpoint of a virtual paired-port setup, and configure the other application for its matching peer.

157. [Intermediate] How do you test both directions?

Send a distinctive string from A to B and verify it at B, then send a different string from B to A. Disable local echo when it could hide a broken link.

158. [Intermediate] What does a virtual serial-port pair do?

It presents two software COM endpoints whose transmitted bytes are delivered to the other endpoint. Each application opens its assigned side; the pair stands in for a cable connection.

159. [Intermediate] Why might a COM port be unavailable?

Another process may already have it open, the chosen port may not exist, or the driver or virtual-pair setup may be missing. Check availability before changing circuit wiring.

160. [Intermediate] What is hardware flow control?

Extra control signals indicate whether a peer can accept more data. RTS/CTS is a common pair; both endpoints and cable wiring must support the selected convention.

161. [Intermediate] What is software flow control?

Special characters, commonly XON and XOFF, request that transmission resume or pause. If arbitrary binary data contains these values, the protocol must handle them appropriately or disable that mode.

162. [Intermediate] Why can a three-wire cable appear correctly wired but transmit nothing?

An endpoint may be waiting for a handshake signal not present on the cable. For the simple lab demonstration, use matching no-flow-control settings if the experiment permits it.

163. [Expert] Is there one universal full-handshake null-modem cable?

No. Required crossings or loopbacks depend on the equipment's control-signal expectations. The three-wire arrangement is enough only when hardware handshaking is unnecessary or appropriately disabled.

164. [Expert] How is loopback different from null modem?

Loopback returns one endpoint's own transmit signal to its receiver. Null modem connects two endpoints. A successful loopback helps isolate a local fault but does not prove the remote cable and endpoint work.

165. [Expert] Why must you inspect COMPIM signal direction rather than blindly connect identical labels?

Connector labels may follow DTE naming while simulator pins represent signals entering or leaving the model. Trace each source to a compatible receiver using the model's pin direction and voltage convention.

12 / Practical: network topologies

166. [Basic] What is network topology?

The arrangement of network devices and links. Physical topology describes connections; logical topology describes how communication flows over them.

167. [Basic] Describe a star topology.

Each endpoint connects to a central device. A failed endpoint cable usually affects one station, while failure of the central device can affect the whole star.

168. [Basic] Describe bus and ring topologies.

A traditional bus shares one backbone medium. A ring connects nodes in a closed loop. Their actual access and fault behavior depend on the link technology and protocol.

169. [Basic] What is a mesh topology?

A topology with multiple interconnections between nodes. A full mesh connects every pair; a partial mesh connects only selected pairs.

170. [Basic] What are tree and hybrid topologies?

A tree uses a hierarchical branching arrangement. A hybrid combines multiple topology patterns in one network.

171. [Intermediate] What is the basic Packet Tracer test in the notes?

Connect PCs and network devices, assign compatible unique IP addresses, then test delivery using ping or a simple PDU. Observe link state and the path taken.

172. [Intermediate] Give a suitable four-PC IPv4 example.

Use 192.168.10.1 through 192.168.10.4 with mask 255.255.255.0 on a common LAN. No default gateway is needed just to communicate within that same subnet.

173. [Intermediate] Which Ethernet cable choices are traditionally taught?

Straight-through for PC-to-switch or PC-to-hub; crossover for similar interfaces such as switch-to-switch when auto-MDI/MDIX is absent. Modern equipment may automatically adjust, so cable rules depend on device support.

174. [Intermediate] What is the difference between a hub and a switch?

A hub repeats signals to other ports. A switch learns source MAC addresses and forwards frames based on destination MAC addresses, reducing unnecessary forwarding for known unicast traffic.

175. [Intermediate] Why can the first ping fail and later pings succeed?

Address resolution or link/topology convergence may still be occurring. Inspect ARP and link state, wait for settling, and retest; do not assume every first failure is harmless.

176. [Intermediate] How many links are needed for a full mesh of n nodes?

n(n-1)/2 undirected point-to-point links. For five nodes, 5 x 4/2 = 10 links.

177. [Intermediate] Is a chain of Ethernet switches a true shared bus?

No. It is a line or chain of point-to-point switched links. It may resemble a bus drawing, but does not reproduce a single shared-coax collision domain.

178. [Expert] What happens if you connect Ethernet switches in a ring?

You create redundant Layer-2 paths. Spanning Tree can block a path to prevent forwarding loops; without suitable loop prevention, broadcasts can circulate and cause a storm.

179. [Expert] Does a physical switch ring implement token-ring networking?

No. Physical shape does not supply a token-passing protocol. Ethernet with a ring of switches still follows Ethernet forwarding and loop-prevention rules.

180. [Expert] Why is an amber or blocked redundant link not necessarily a fault?

Spanning Tree may intentionally place a redundant port in a non-forwarding state. Check protocol state and end-to-end reachability before attempting to force every link to forward.

13 / IP addressing and subnetting

181. [Basic] How large is an IPv4 address?

32 bits, commonly written as four decimal octets from 0 to 255. An address and prefix length together identify the network and host portions.

182. [Basic] How large is an IPv6 address?

128 bits, normally written in hexadecimal groups separated by colons. IPv6 expands the address space and differs from IPv4 in several protocol details.

183. [Basic] What is a subnet mask?

A 32-bit value whose contiguous leading ones identify the network prefix in ordinary CIDR addressing. For example, /24 corresponds to 255.255.255.0.

184. [Basic] What is a default gateway?

The next-hop router used when no more specific route applies, typically to reach destinations outside the local subnet. It must be reachable on the local link.

185. [Basic] What is ping?

A diagnostic that commonly sends ICMP Echo Requests and receives Echo Replies. A reply demonstrates IP reachability for that exchange, not that every application service works.

186. [Intermediate] Find network, broadcast and hosts for 192.168.10.25/24.

Network: 192.168.10.0. Broadcast: 192.168.10.255. Ordinary usable host range: 192.168.10.1-192.168.10.254, giving 254 hosts.

187. [Intermediate] Find the subnet for 192.168.10.70/26.

/26 is 255.255.255.192, giving blocks of 64. Address 70 is in network 192.168.10.64, broadcast .127, with ordinary hosts .65-.126.

188. [Intermediate] What are the private IPv4 ranges?

10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16. They are intended for private addressing and are not globally routed as ordinary public addresses.

189. [Intermediate] How does a host determine whether a destination is local?

It applies the relevant prefix/mask to its own address and the destination. Matching local subnet prefixes normally allow direct delivery; otherwise routing via a next hop is needed.

190. [Intermediate] What does ARP do?

On IPv4 Ethernet, it resolves a local next-hop IPv4 address to a MAC address. For a remote destination the host usually resolves the gateway's MAC, not the remote host's MAC.

191. [Intermediate] What do DHCP and DNS do?

DHCP can supply address configuration such as IP, mask and gateway. DNS resolves names to records such as IP addresses. They solve different problems.

192. [Intermediate] Why can duplicate IP addresses break a lab?

Different machines may answer for the same address, causing inconsistent address resolution and delivery. Give each host a unique usable address within the intended subnet.

193. [Expert] Why is '2^h - 2 hosts' not universal?

It applies to conventional IPv4 subnets reserving all-zero host bits for the network and all-one host bits for broadcast. /31 point-to-point addressing and /32 routes are special cases; do not apply the rule blindly.

194. [Expert] Are class A, B and C masks how all networks are routed today?

No. They are useful historical classful concepts. CIDR uses explicit prefix lengths, so a 192.x address need not be configured as /24.

195. [Expert] Ping by address works but a hostname fails; what next?

Check DNS configuration and name resolution. The successful address test suggests basic reachability, but does not establish that the DNS server or the name record is correct.

14 / OSI, TCP/IP and network devices

196. [Basic] Name the seven OSI layers from bottom to top.

Physical, Data Link, Network, Transport, Session, Presentation and Application. Explain a function of each instead of only memorizing the names.

197. [Basic] What does the physical layer do?

It carries signals representing bits over a medium, including electrical/optical behavior, signaling and physical interface aspects.

198. [Basic] What does the data-link layer do?

It handles frames across a link, including link addressing, medium access and error detection as provided by the particular link technology.

199. [Basic] What does the network layer do?

It handles logical addressing and forwarding across networks. IP operates here; routers choose next hops using routing information.

200. [Basic] What does the transport layer do?

It supports communication between processes on hosts. TCP provides reliable ordered byte-stream delivery, while UDP provides datagrams with minimal transport services.

201. [Intermediate] How does the TCP/IP model map to OSI?

A common four-layer model uses Link, Internet, Transport and Application. Its Application layer covers much of OSI's upper three layers; educational five-layer variants split Physical from Data Link.

202. [Intermediate] What is encapsulation?

Each layer adds control information around the higher-layer data. For example, application bytes may be carried in TCP, then IP, then an Ethernet frame.

203. [Intermediate] Distinguish MAC address, IP address and port number.

A MAC address identifies a link-layer interface in the local forwarding context; an IP address supports network-layer delivery; a transport port identifies an endpoint service or conversation on a host.

204. [Intermediate] Compare bridge, switch and router.

Bridges and ordinary Layer-2 switches forward using MAC addresses. Routers forward between IP networks. A multilayer switch can also perform routing, so identify the function being used.

205. [Intermediate] Why does a switch learn source MAC addresses?

The arrival port tells it where the source is reachable. It uses that learned table later to choose an output port for frames addressed to that source.

206. [Intermediate] What happens to an unknown unicast at a typical switch?

It is flooded within the relevant VLAN to eligible ports except the incoming port. Once the destination's location is learned, known unicast can be forwarded selectively.

207. [Intermediate] What is a broadcast domain?

A set of interfaces that receive a link-layer broadcast. A VLAN commonly forms one broadcast domain; routers normally separate such domains.

208. [Expert] Does UDP mean an application cannot be reliable?

No. UDP itself does not provide TCP-style retransmission and ordering, but an application or protocol above it can implement reliability, sequencing and congestion control.

209. [Expert] Why does a router change Ethernet addresses but not normally the destination IP?

Each hop needs a new local-link frame addressed to its next hop. The IP destination normally identifies the end destination, except when mechanisms such as NAT intentionally rewrite it.

210. [Expert] Does a successful ping prove an HTTP server is working?

No. ICMP reachability and TCP/application availability are different tests. The service may be stopped or filtered even when ping replies succeed.

15 / Transmission media and impairments

211. [Basic] What are guided and unguided media?

Guided media constrain signals to a physical path such as copper or fiber. Unguided media use electromagnetic propagation through space, such as radio and microwave links.

212. [Basic] Why are wires twisted in twisted-pair cable?

Twisting helps balance interference pickup and reduce crosstalk between pairs. Performance also depends on pair geometry, termination and the signaling method.

213. [Basic] What is coaxial cable?

A cable with a central conductor, dielectric, surrounding conductor/shield and outer insulation. Its controlled geometry supports a defined impedance and shielding.

214. [Basic] How does optical fiber carry information?

Light propagates along a dielectric waveguide. A transmitter modulates an optical signal and a receiver detects it; the cable does not carry the data as an electrical current.

215. [Basic] What is crosstalk?

Unwanted coupling from one signal path into another. It can occur between copper pairs and reduces the receiver's signal quality.

216. [Intermediate] Compare single-mode and multimode fiber.

Single-mode fiber supports one principal spatial mode over its operating range; multimode fiber supports multiple modes and can suffer modal dispersion. Actual reach depends on the fiber and transceivers.

217. [Intermediate] What is dispersion?

Pulse spreading because propagation differs by mode or wavelength. Spreading can cause neighboring pulses to overlap and limit usable data rate or distance.

218. [Intermediate] Compare a repeater with an amplifier.

An amplifier boosts a waveform including its noise. A digital regenerative repeater can decide symbols and recreate timing and signal levels, provided the incoming signal is still recoverable.

219. [Intermediate] Why is fiber resistant to electromagnetic interference?

Information travels optically through a dielectric medium rather than by an electrical signal in a conductor. End equipment can still be affected by electrical faults or interference.

220. [Intermediate] Why do terrestrial microwave links often require line of sight?

Their directional high-frequency propagation is strongly affected by obstructions and path geometry. Antenna alignment and adequate clearance are therefore important.

221. [Intermediate] What is a satellite transponder?

Equipment that receives an uplink signal and retransmits it on a downlink, often with filtering, amplification and frequency conversion. Its exact processing depends on the satellite design.

222. [Intermediate] Why can a satellite link have large delay?

Signals travel a long physical path. Geostationary systems in particular have long propagation distances; satellite orbit and routing determine the actual delay.

223. [Expert] Can stronger amplification fully remove transmission errors?

No. Amplification cannot reconstruct information already lost in noise or distortion and may cause clipping. Equalization, coding, regeneration or a better channel may be needed.

224. [Expert] Why is shielded cable not automatically the best installation?

Shield termination, grounding, connector quality, cost and environment matter. Poor installation can negate the intended interference benefit.

225. [Expert] Why does a published cable distance alone not determine link performance?

Data rate, loss, capacitance, bandwidth, termination, transceiver capability and interference all matter. Use a standard or device specification together with the actual installation conditions.

16 / Line coding and scrambling

226. [Basic] What is line coding?

Mapping digital data to a baseband signal pattern suitable for transmission. It differs from carrier modulation such as ASK and from error-control coding such as Hamming coding.

227. [Basic] What is NRZ-L?

Non-Return-to-Zero Level encoding represents bits using two signal levels held throughout the bit interval. Which level means 0 or 1 must be declared.

228. [Basic] What is NRZ-I?

Non-Return-to-Zero Invert encoding represents one binary value by a transition at the start of its interval and the other by no transition. State the convention and initial level.

229. [Basic] What is Manchester encoding?

A line code with a transition in the middle of every bit. The direction of that transition represents the bit under a stated convention and also provides timing information.

230. [Basic] What is AMI?

Alternate Mark Inversion represents zeros by zero level and successive ones by alternating positive and negative pulses. It is a bipolar three-level line code.

231. [Intermediate] Why can a long run of identical NRZ bits be troublesome?

Few or no transitions make receiver clock recovery difficult. Certain patterns also create DC or low-frequency components that a channel may not pass well.

232. [Intermediate] How does differential Manchester encode information?

It always has a mid-bit transition for timing; the presence or absence of a transition at the bit boundary encodes the bit under the chosen convention. It relies on transitions rather than absolute polarity.

233. [Intermediate] What is the purpose of scrambling in AMI systems?

Replace problematic runs, particularly long zero runs, with recognizable patterns containing transitions. It helps timing while allowing the receiver to reconstruct the original data.

234. [Intermediate] What does B8ZS replace?

A run of eight zeros is replaced by 000VB0VB, where V is a deliberate bipolar violation and B is a normal bipolar pulse. Actual pulse signs depend on the preceding nonzero pulse.

235. [Intermediate] What does HDB3 replace?

A run of four zeros is replaced by 000V or B00V. The choice depends on the parity of nonzero pulses since the previous violation, preserving useful timing and DC balance.

236. [Intermediate] What do B and V mean?

B obeys ordinary AMI alternation relative to the preceding nonzero pulse. V deliberately violates it by repeating that polarity, allowing the receiver to recognize a substitution.

237. [Intermediate] Does scrambling automatically encrypt data?

No. The substitution rules are designed for signal properties and are reversible by the receiver. They do not by themselves provide cryptographic confidentiality.

238. [Expert] Why must an examiner be told your Manchester convention?

Textbooks and standards can assign opposite transition directions to 0 and 1. A waveform is not meaningfully judged until the convention and timing boundaries are specified.

239. [Expert] Is a bipolar violation always a transmission error?

Not in a deliberately scrambled code such as B8ZS or HDB3. A recognized substitution contains planned violations; an unexpected violation may indicate an error.

240. [Expert] Why can a line code with more transitions require more bandwidth?

Faster waveform changes introduce higher-frequency content. Timing recovery improves, but the required spectrum depends on pulse shape and how bandwidth is defined.

17 / Sampling, PCM and delta modulation

241. [Basic] What is sampling?

Measuring a continuous-time signal at discrete times. The samples still have continuous-valued amplitude until quantization is applied.

242. [Basic] What is quantization?

Mapping each sample amplitude to one of a finite set of levels. It introduces an amplitude error; it is not simply the deletion of redundant binary bits.

243. [Basic] What is PCM?

Pulse Code Modulation represents sampled and quantized amplitudes as binary words. Its basic operations are sampling, quantization and binary encoding.

244. [Basic] What is aliasing?

Different analog frequency components become indistinguishable after inadequate sampling. An anti-alias low-pass filter removes unwanted components before the sampler.

245. [Basic] What is delta modulation?

A scheme that represents the direction of change of a staircase approximation, commonly with one up/down bit per sample and a fixed step size.

246. [Intermediate] State the sampling requirement for a low-pass band-limited signal.

The sampling frequency must exceed twice the highest relevant signal frequency for practical reconstruction with suitable filtering. The textbook ideal limiting rate is 2 fmax.

247. [Intermediate] What is the textbook minimum sampling rate for a 200 kHz low-pass signal?

400,000 samples/s is the ideal Nyquist limit. A practical design uses margin and an anti-alias filter rather than assuming arbitrary signals are safely captured exactly at the limit.

248. [Intermediate] Find PCM bit rate for 8000 samples/s and 8 bits/sample.

8000 x 8 = 64,000 bit/s, or 64 kbit/s, before framing, synchronization or error-control overhead.

249. [Intermediate] How many levels are available with n bits per sample?

2^n. Eight bits provide 256 quantization levels. More levels can reduce quantization error for a fixed amplitude range.

250. [Intermediate] State the ideal uniform-quantizer SNR approximation.

For a full-scale sine wave and ideal n-bit quantization, SQNR is approximately 6.02n + 1.76 dB. This is not a universal SNR formula for any waveform or any noise source.

251. [Intermediate] What is slope-overload distortion?

In fixed-step delta modulation, the staircase cannot change fast enough to follow the input slope. Increasing step size or sampling rate can help.

252. [Intermediate] What is granular noise?

The delta-modulator staircase repeatedly overshoots a slowly varying input when its step is too large. Reducing the step helps but can worsen slope overload for fast changes.

253. [Expert] Why use adaptive delta modulation?

It changes the step size according to signal behavior, using larger steps for steep changes and smaller steps for slowly varying regions. This addresses the fixed-step compromise.

254. [Expert] Does one bit per sample guarantee lower bit rate than PCM?

No. Delta modulation may require much higher sampling frequency. Compare total bits per second and achieved quality, not just bits per sample.

255. [Expert] How many ideal bits are needed for at least 40 dB full-scale-sine quantization SNR?

n >= (40-1.76)/6.02, about 6.35, so at least 7 bits. Real converter noise and distortion can require additional nominal resolution.

18 / Multiplexing and transmission timing

256. [Basic] What is multiplexing in communication systems?

Combining multiple information streams for transport over a shared link while keeping them separable at the receiver.

257. [Basic] What is FDM?

Frequency Division Multiplexing assigns channels different frequency bands so they can occupy the link at the same time.

258. [Basic] What is TDM?

Time Division Multiplexing assigns channels separate time slots on a shared link. The receiver uses timing and framing to recover the channels.

259. [Basic] What is WDM?

Wavelength Division Multiplexing carries multiple optical channels using different wavelengths on one fiber. It is the optical counterpart of separating channels by frequency.

260. [Basic] What is a guard band?

A frequency gap between channels used to reduce interference when filters and spectra are not ideal. It consumes spectrum without directly carrying payload.

261. [Intermediate] Find FDM bandwidth for five 100 kHz channels with 10 kHz gaps between adjacent channels.

There are four internal gaps, so bandwidth is 5 x 100 + 4 x 10 = 540 kHz, excluding any additional outer guard bands.

262. [Intermediate] Compare synchronous and statistical TDM.

Synchronous TDM assigns recurring slots even to idle sources. Statistical TDM allocates capacity to active sources and typically needs channel identification and buffering.

263. [Intermediate] Four 1 kbit/s inputs are bit-interleaved without overhead. What is output rate?

4 kbit/s. Each frame contains one bit from each input, so it has four bits and repeats 1000 times per second.

264. [Intermediate] Add one framing bit to each of those frames. What happens?

Each frame now has five bits at 1000 frames/s, so line rate is 5 kbit/s and payload efficiency is 4/5 = 80 percent.

265. [Intermediate] Compare synchronous and asynchronous serial transmission.

Synchronous transmission shares or recovers timing for a continuous stream or block. Asynchronous character transmission uses start/stop framing and allows gaps; it still requires compatible bit timing within each frame.

266. [Intermediate] What is an isochronous requirement?

Delivery with timing suitable for a time-sensitive stream, such as audio/video. It concerns regular timing or bounded delay variation and should not be confused with loss-free delivery.

267. [Intermediate] How does QPSK help carry data at a given symbol rate?

Four phase states carry two bits per symbol. For the same gross bit rate and comparable pulse shaping, its symbol rate is half that of BPSK.

268. [Expert] Is a 1 Mbit/s 16-QAM stream guaranteed to fit inside 250 kHz?

It has 250 ksymbols/s because each symbol carries four bits. Occupied bandwidth also depends on pulse shaping, roll-off, filtering and guard bands; equating baud directly to hertz needs stated assumptions.

269. [Expert] Why must a TDM receiver synchronize to frame boundaries?

Correct bit timing alone does not identify which channel owns a slot. A frame slip can deliver valid-looking bits to the wrong output channel.

270. [Expert] What happens when statistical TDM sources all become busy?

The aggregate offered load may exceed link capacity. Buffers grow, delay rises, and data can be dropped or backpressured; statistical sharing does not create unlimited capacity.

19 / Error control, flow control, ARQ and HDLC

271. [Basic] What is error detection?

Checking received data for inconsistency using redundancy such as parity, checksum or CRC. Detection alone does not establish which bits to repair.

272. [Basic] What is a burst error?

Corruption extending from the first affected bit to the last within a span. Not every bit inside that span must be wrong.

273. [Basic] What is flow control?

Managing how much data a sender transmits so the receiver is not overwhelmed. It differs from error control, which addresses damaged, lost or duplicated data.

274. [Basic] What is ARQ?

Automatic Repeat reQuest uses acknowledgments, sequence numbers and timers to recover through retransmission. It needs a return path or another acknowledgment mechanism.

275. [Basic] What is stop-and-wait ARQ?

Send one frame and wait for acknowledgment before advancing. On timeout retransmit; sequence numbers let the receiver recognize duplicates.

276. [Intermediate] How do Go-Back-N and Selective Repeat differ?

Go-Back-N retransmits from an unacknowledged frame onward and normally discards out-of-order arrivals. Selective Repeat buffers suitable out-of-order frames and retransmits the missing or damaged ones individually.

277. [Intermediate] Why are sequence numbers needed?

They distinguish new data from duplicate retransmissions, especially when an acknowledgment is lost. A correct receiver should not deliver the same data twice just because the sender timed out.

278. [Intermediate] What is the window limit with three sequence bits?

In the usual protocols, Go-Back-N allows sender window at most 2^3-1 = 7; Selective Repeat with equal sender/receiver windows allows at most 2^(3-1) = 4 to avoid ambiguity after wraparound.

279. [Intermediate] What is a CRC?

A polynomial-based error check computed by modulo-2 division. The sender appends a remainder; the receiver checks the resulting codeword using the same generator and conventions.

280. [Intermediate] Why is XOR used in CRC division?

Addition and subtraction over binary polynomial coefficients are both XOR, with no carries or borrows. This is different from ordinary integer division.

281. [Intermediate] What are HDLC's flag and stuffing rule?

The flag is 01111110. Within the frame content, insert a zero after five consecutive ones; the receiver removes such stuffed zeros. Flags themselves are delimiters, not stuffed payload.

282. [Intermediate] What are HDLC I, S and U frames?

Information frames carry user data and control information; Supervisory frames support flow/error control; Unnumbered frames support link management and other control functions. The control field identifies the type.

283. [Expert] Why does stop-and-wait perform poorly on a long-delay link?

The sender is idle while waiting. With negligible ACK transmission time and no errors, utilization is approximately 1/(1+2a), where a is propagation time divided by frame transmission time.

284. [Expert] Can a zero CRC remainder prove there was no corruption?

No. Some error patterns are multiples of the generator and remain undetected. The generator and message length determine detection guarantees; a CRC is also not a cryptographic authenticity check.

285. [Expert] How does parity compare with Hamming and CRC?

A single parity bit detects odd-count bit flips but cannot locate one. Hamming adds structured checks for single-error correction. CRC is designed for strong detection, including specified burst-error guarantees, and is usually paired with retransmission rather than used to locate errors.

20 / Switching, legacy networks and expert synthesis

286. [Basic] What is circuit switching?

Establishing a reserved end-to-end communication path or allocation for a session. Setup, data transfer and release are the usual phases.

287. [Basic] What is packet switching?

Dividing data into packets that share network resources with other traffic. Packets may experience queueing, loss or different routes depending on the service.

288. [Basic] What is message switching?

Store-and-forward handling of an entire message at each intermediate node. It can require substantial storage and introduce long delay compared with forwarding smaller packets.

289. [Basic] What is a datagram network?

A connectionless packet network where packets are forwarded independently using destination information. IP provides a datagram service.

290. [Basic] What is a virtual circuit?

A logical connection established through a packet network, with forwarding state or identifiers along a chosen path. It is not necessarily a dedicated physical circuit.

291. [Intermediate] Compare virtual-circuit and datagram delivery.

Virtual-circuit service establishes connection state before or as part of use and identifies traffic by a circuit/connection label. Datagram service does not require such per-connection setup for basic forwarding.

292. [Intermediate] What is ATM and what is its cell size?

Asynchronous Transfer Mode is a connection-oriented cell-switching technology. Each cell has 53 bytes: a 5-byte header and 48-byte payload.

293. [Intermediate] Why did ATM use small fixed-length cells?

Fixed short units simplify switching and bound the serialization delay of one cell, helping mixed voice/data traffic. The design incurs header overhead and adaptation complexity.

294. [Intermediate] What are SONET and SDH?

Standardized synchronous transport hierarchies widely associated with optical networks. They organize rates, framing and multiplexing; they are more than simple electrical-to-optical conversion.

295. [Intermediate] What is DSL?

Digital Subscriber Line is a family of technologies carrying digital communication over copper subscriber loops. Reach and data rate depend on the DSL type, line condition and interference.

296. [Intermediate] What is ISDN?

Integrated Services Digital Network, a digital telecommunications service family supporting integrated voice/data services. A common basic-rate interface is described as 2B+D; it is a legacy syllabus topic here.

297. [Intermediate] What should you say if asked which modulation is 'best'?

Ask what matters: required bit rate, bandwidth, power, channel noise, receiver complexity or phase synchronization. ASK, FSK and PSK have different tradeoffs; no scheme is universally best.

298. [Expert] How would you troubleshoot any failed lab systematically?

Confirm the intended input and expected output, then inspect power/reference, connections, component function, settings and measurement points. Test one block at a time with a simple known input and change one variable per trial.

299. [Expert] Why is a plausible screenshot insufficient proof of a correct experiment?

One input can accidentally produce the expected output in a faulty circuit. Demonstrate multiple cases, including boundary and fault cases, and connect each observation to a truth table, equation or known message.

300. [Expert] How should you answer when the notebook conflicts with the circuit?

State the standard principle, identify the specific mismatch, and verify the actual component or configuration. For example: 'The two FSK tones must differ; I would check both generator frequencies before accepting this output.'

Extra course terms to recognise

Term	Viva reminder
HDLC modes	NRM: secondary sends when polled. ARM: secondary may initiate without explicit permission. ABM: balanced combined stations can initiate.
HDLC supervisory frames	RR: receive ready; RNR: receive not ready; REJ: reject/retransmission request; SREJ: selective reject where supported.
X.25	A connection-oriented packet-switched interface/protocol suite with physical, link and packet layers; LAPB is its link-layer protocol. It is a legacy course topic.
Frame Relay	A legacy frame-switching service with less link-by-link error recovery than X.25; higher layers are expected to handle recovery.
Interleaving	Rearranges coded symbols so a burst may spread across several codewords, helping a code designed for limited errors per word.
Bandwidth-delay product	Bit rate multiplied by delay gives the number of bits that can occupy a path over that interval. State whether the delay is one-way or round trip.

Last-minute sheet | Logic and Hamming

S1 S0	MUX Y	DEMUX when D=1
00	D0	Y0=1, others=0
01	D1	Y1=1, others=0
10	D2	Y2=1, others=0
11	D3	Y3=1, others=0

DEMUX when D=0: all outputs=0. These are active-high conventions. MUX: four AND terms joined by OR. DEMUX: four AND outputs kept separate.

Position	1	2	3	4	5	6	7
Role	P1	P2	D1	P4	D2	D3	D4
Data 1011	0	1	1	0	0	1	1
Error at 5	0	1	1	0	1	1	1

P1=D1 XOR D2 XOR D4; P2=D1 XOR D3 XOR D4; P4=D2 XOR D3 XOR D4.

S1 checks 1,3,5,7; S2 checks 2,3,6,7; S4 checks 4,5,6,7.

Read syndrome as S4 S2 S1: 001->1, 010->2, 011->3, 100->4, 101->5, 110->6, 111->7.

Correct only under the single-error assumption. Syndrome zero means no detected error, not proof against all multi-bit errors.

Extended Hamming: up to two errors

Syndrome S	Overall parity check P	Decision
0	0	No detected error
Nonzero	1	Correct bit at position S
0	1	Error in the added overall parity bit
Nonzero	0	Double error detected; do not correct

Overall parity check P is XOR of all eight received bits under even parity. This table is not a guarantee for three or more errors.

Last-minute sheet | Serial and networks

Item	Remember
Serial framing	8N1: 1 start + 8 data + 0 parity + 1 stop = 10 bits/character
9600 bit/s	960 characters/s maximum with continuous 8N1 framing
Standard DTE DE-9	2=RX, 3=TX, 5=signal ground
Three-wire null modem	A3-B2, A2-B3, A5-B5; no unsupported hardware-handshake requirement
Loopback	Returns one endpoint's TX to its own RX
MAX232	Logic/RS-232 level conversion, not UART framing
RS-232 data idle	Mark / logical 1 / negative line voltage
Same subnet	Unique addresses and matching prefix; no gateway needed just for local peers
Local example	192.168.10.1-.4 /24; mask 255.255.255.0
Switch loops	STP can block a redundant path; a blocked port can be normal

Fault -> first checks

Symptom	First checks
No serial characters	Port exists and is free; TX-to-RX paths; compatible levels; flow control
Gibberish	Baud rate, data bits, parity, stop bits, polarity
Double characters	Local echo plus remote echo or loopback
Ping fails	Cable/link state, IP/mask, duplicate IP, VLAN and next-hop route
Modulation trace unclear	Input data, generator settings, probe point, trigger and time scale

Worked numerical practice

1. Symbol rate

A 16-QAM signal carries log2(16)=4 bits/symbol. At 2400 baud its gross data rate is 9600 bit/s, before coding and framing overhead.

2. SNR and capacity

S=10 mW and N=1 microW gives SNR=10,000 and 40 dB. If B=3 kHz, Shannon capacity is 3000 log2(10001), approximately 39.86 kbit/s. Do not substitute 40 directly as the linear SNR.

3. Sampling and PCM

For a textbook 4 kHz low-pass voice signal, the ideal limiting sample rate is 8 ksample/s. At 8 bits/sample, raw PCM is 64 kbit/s. Real filters and overhead require design allowances.

4. Quantizer levels

For an ideal 8-bit uniform quantizer spanning 0-5 V under a 256-bin convention, step size is 5/256 = 19.531 mV. Ideal rounding error is at most half a step within range; clipping is a different error.

5. Delay

A 1000-byte frame has 8000 bits. At 1 Mbit/s, transmission takes 8 ms. Over 200 km at 2 x 10^8 m/s, one-way propagation is 1 ms. Total without queue/processing is 9 ms.

6. Stop-and-wait utilization

With frame transmission 1 ms and one-way propagation 10 ms, a=10. Ignoring errors and ACK transmission, utilization is 1/(1+20)=1/21, about 4.76 percent. A larger sliding window can keep the link busy.

7. Serial transfer time

Sending 100 ASCII characters at 9600 bit/s in 8N1 requires at least 100 x 10/9600 = 0.1042 s, excluding gaps and waiting.

8. Hamming parity count

For m=8 data bits, r=3 fails because 8 < 8+3+1. r=4 works because 16 >= 8+4+1. A shortened single-error-correcting construction uses 12 total bits.

9. Subnetting

192.168.10.70/26: mask 255.255.255.192; block size 64; network .64; broadcast .127; conventional host range .65-.126; 62 usable hosts.

10. FDM and TDM

Five 100 kHz channels with four 10 kHz internal guard bands need 540 kHz. Three 3 kbit/s bit-interleaved TDM inputs need 9 kbit/s without overhead, frame period 1/3000 s and slot period 1/9000 s.

CRC example and common oral challenges

Modulo-2 CRC example

Illustrative data=1101, generator=1011 (degree 3). Append three zeros to obtain 1101000. Divide by XOR: the remainder is 001. Transmit 1101001. Dividing this codeword by 1011 produces zero remainder. This example uses no reflection, initial register preset or final XOR.

Long-division steps: 1101 XOR 1011 = 0110; bring down 0 -> 1100; 1100 XOR 1011 = 0111; bring down 0 -> 1110; 1110 XOR 1011 = 0101; bring down 0 -> 1010; 1010 XOR 1011 = 0001; keep the final three bits -> 001.

Rehearse these without notes

1. Explain why your ASK output is not FSK.
2. Prove that the FSK sources have different frequencies.
3. Show the 180-degree BPSK relationship using a reference.
4. Derive the 4-to-1 MUX equation from its selection table.
5. Explain why DEMUX outputs are all zero when D=0.
6. Encode 1011 using a declared Hamming convention.
7. Inject an error at position 5 and show the syndrome.
8. Explain why two errors can fool a single-error corrector.
9. Explain MAX232 without calling it a microcontroller or UART.
10. Trace the complete serial signal path from sender to receiver.
11. Show why local echo alone is not a communication test.
12. Explain null modem, loopback and DTE/DCE.
13. Give a valid four-PC subnet and test it.
14. Explain an STP-blocked link in a physical switch ring.
15. Calculate bit rate, baud rate and serial character rate without mixing them.

The five-part answer structure

Aim -> components -> operating principle -> expected observation -> verification/fault check. Keep the first answer to about 20-40 seconds, then expand when the examiner asks a follow-up.

Sources and verification notes

Uploaded practicals define the experiment scope: 2-1. DT.pdf (28 pages) and 2-2. DT.pdf (44 pages). Both are scanned handwritten notebooks. The scope table maps each practical to its PDF page positions.

Course theory: D_T 01-13 and IP Address.pdf from slide.zip, plus Introduction, Data Transmission, Data Encoding, DC Techniques, Data Link Control and Multiplexing from the DT chapter-note archive. Scanned text was read with OCR and selected circuit pages were inspected visually; unclear wiring is explicitly identified.

These are original study explanations, not a transcription of every slide. Instructor-specific wiring and simulator models still need to match the demonstrated lab setup. The reference loopback and numerical settings are labelled examples, not claims that the notebook used those exact settings.

Texas Instruments: CD4016B bilateral analog switches

Texas Instruments: SN74LS04 hex inverters

Texas Instruments: LF398 sample-and-hold family

Texas Instruments: CD4073 three-input AND gates

Texas Instruments: SN74LS138 decoder datasheet

Texas Instruments: MAX232 driver/receiver datasheet

Analog Devices: Fundamentals of RS-232 serial communications

MIT: Matrix Hamming Codes

Cisco: Spanning Tree Protocol

Validation: all 16 four-bit inputs were encoded and all 112 single-bit corrupted words checked for correct syndrome and restoration. MUX selection was checked for all 64 data/select combinations. Numerical examples were independently calculated. No Proteus circuit or Packet Tracer project was executed here; the guide does not certify the scanned schematics as tested hardware.